

PROJECT SYNOPSIS

NAME: DATTA DHEERAJENDRA KV, KUDUM KUSUMA

USN: P19MT24S126052, P19MT24S126048

COLLEGE: EAST POINT COLLEGE OF HIGHER EDUCATION

COURSE: MASTER OF COMPUTER APPLICATION (MCA)

SEMESTER: 3RD SEMESTER

Project Title: AI Based Personalized Investment Recommendation System Using Machine Learning

Introduction:

With the rapid growth of digital financial services, individuals increasingly seek intelligent tools to manage and grow their investments. However, most investors lack the knowledge required to select appropriate investment options according to their financial goals and risk tolerance. Traditional financial advisory services are often expensive or inaccessible for many users. This project focuses on developing an AI based system that analyzes user financial data and investment preferences to recommend suitable investment strategies using machine learning techniques.

Problem Statement:

Many individuals invest without proper knowledge of financial markets, risk levels, or portfolio diversification. Existing investment platforms provide general suggestions but lack personalization based on a user's income, savings pattern, financial goals, and risk tolerance. This results in inefficient investment decisions and potential financial losses.

Objectives:

- To analyze user financial data such as income, savings, expenses, and risk tolerance.
- To classify investors into different risk categories using machine learning techniques.
- To recommend personalized investment portfolios based on user profiles.
- To improve financial decision making and promote smarter investment planning.

Methodology:

The proposed system collects user financial details including income level, savings behavior, investment goals, and risk appetite. Data preprocessing and exploratory analysis are performed to understand user financial patterns. Machine learning algorithms such as K-Means clustering are used to group users based on their risk profiles. A recommendation model is then designed to suggest suitable investment options such as low risk, moderate risk, or high growth portfolios.

The system continuously improves recommendations using user feedback and updated financial data

Tools and Technologies:

Python, Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow, and Visual Studio Code

Expected Outcome:

The proposed system will provide intelligent and personalized investment recommendations to users based on their financial profile. It will help individuals understand their investment behavior, choose appropriate financial instruments, and manage their money more effectively. This project demonstrates the practical application of artificial intelligence and machine learning in the financial technology (Fin Tech) domain

